Patellar and Quadricep Tendinopathy

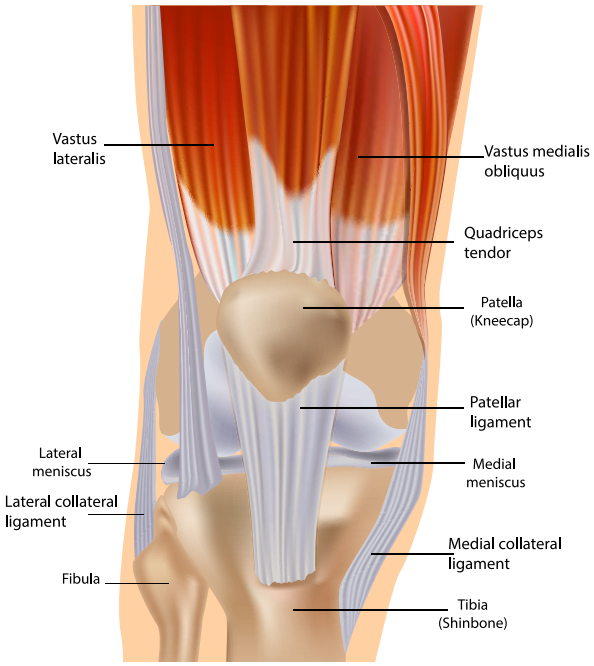
Background

- Tendinopathy of the patella (your kneecap) and quadricep (large muscle in the front of your thigh) is caused by overuse of the patellar and quadricep tendons. It can affect both athletes and non-athletes alike. Additional causes can be inadequate recovery and poor biomechanics.
- The quad tendon connects the quad muscle group to the upper portion of the patella . The patellar tendon attaches the bottom portion of the kneecap to the lower leg at the bony prominence called the "tibial tuberosity."
- Tendinopathy symptoms included pain with use (such as jumping or running), mild swelling, and tenderness to the touch.
- Diagnosis is made through review of your history and an exam that commonly includes an ultrasound and X-rays. Damage seen within the tendon confirms the diagnosis.
- Treatment involves initial rest and activity modification, as well as rehabilitative exercises to strengthen the tendons and improve pain.
- During rehab, progressive loading (very gradually increasing weight) on a consistent basis (6+ days per week) is crucial to achieve positive results.

Latest Research

- The use of therapeutic injections might be helpful for treatment, especially if physical therapy hasn't provided sufficient improvement.
- Injection therapies include dextrose prolotherapy (injection of sugar water and lidocaine), platelet-rich plasma (PRP), and percutaneous ultrasonic tenotomy (Tenex®).

TENDONS OF THE KNEE





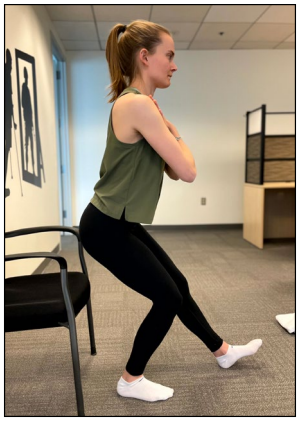
Rehab Instructions

Wall Sits (Isometric)

1. Start with your feet shoulder-width apart and your back against a wall.
2. Slide down the wall until your knees are bent at about 45°.
3. Hold this position for 15–30 seconds.
4. Keep your weight evenly distributed between both legs.
5. To progress, gradually shift more weight to your affected side.

Complete 1 set of 5 reps 1x a day.

Sit to Stand (single leg squat with a chair)


1. Put a chair against a wall for stability.
2. Sit on the edge of the chair with one leg back and the leg other forward.
3. As you stand up, the leg that's staggered back will straighten first.
4. Once you're standing up, slowly lower yourself back onto the chair.
5. Keep your chest up and don't let your back arch or bend.
6. Use a chair or stool with a high seat height to make the exercise easier. Start with what's tolerable!

Complete 2 sets of 20 reps 1x a day. Progress to 3 sets of 20 reps 1x a day.

Box Step Downs




1. Use a stable step or plyo box.
2. Start by stepping onto the box.
3. Step off the box and try to land softly on both feet and bend your knees to absorb the impact.
4. Complete 2 sets of 10 reps and progress to 3 sets of 10 as you can.
5. When your pain improves and you get stronger, add a vertical jump after the landing.
6. **Only perform plyometrics (explosive movements such as jumping or rebounding) 2–3 times per week**. It takes longer to recover from them than from basic weightlifting-type movements.

Complete 2 sets of 10 reps, 1x a day. Progress to 3 sets of 10 reps 1x a day.

Practical Tips for Success

- Performing these exercises regularly will help treat patellar and quad tendinopathy and help prevent it from recurring.
- These exercises require progressive loading to be successful. Progressive loading can include increasing repetitions, adding weight, or increasing the load.
- Mild to moderate pain during the activity is acceptable. **However,** if you experience pain greater than a 3 out of 10 scale back by using less resistance or doing fewer reps (or both) to prevent re-injury.

References

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